


FM-FAM · CHEAT SHEET

# FM-FAM at a glance

For FM-FAM 0.2.5 on the M-VAVE FM-1

DFAM style 16-step sequencer and a drum synth for the FM-1. It has two oscillators with 16 wave types, a kick with its own sequencer, parameter locks, a sound saved per pattern, an effects chain, random pages, scales, presets and projects. Bluetooth audio is experimental.

**FM-FAM is beta software.** Use it at your own risk. It can have bugs and fail (wrong behaviour, lost data, an FM-1 that does not start until you install a firmware again), and installing it or restoring a backup writes to the FM-1's flash. Save what matters first and never unplug the USB cable while it writes. It is provided "as is", without warranty (MPL-2.0, sections 6 and 7); the author is not responsible for any damage to your FM-1, your data or anything connected to it.

`BUTTON` a button, key or encoder of the FM-1 · tap, hold, turn · `A` + `B` hold A while you use B · K1-K4 are the four knobs, in the order of the screen footer · the white keys are the steps

## Getting around

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| `SEQ` `EDIT` `LFO` `ARP` `FX` `GLO` `SCL` `SAVE` | Mode buttons: sequencer, synth, patch bay, kick, effects, patterns and settings, scale, projects and presets. The first tap goes to the mode, on the page you left; each further tap moves to the next page and wraps after the last. `ENV` and `HOME` do nothing alone. |
| `SELECT` turn | Change page inside the mode (does not wrap). |
| `ALGORITHM` turn | Move the cursor of the page: the step (SEQ pages 1-2, kick sequence), the target (patch bay), the bank or slot (patterns), the item (SAVE). Everywhere else it changes the tempo. |
| `PRESETS` turn | Next or previous preset of the mode (synth, kick, FX, patch bay, scale). In patterns it picks the pattern. |
| `KNOB 1`-`KNOB 4` | The four parameters named at the bottom of the screen: green, cyan, magenta and orange. |
| `OCT-` + `KNOB` | Coarse: 8 steps per click on 0-127 knobs, x4 on pitch, velocity, gate, kick pitch and volume, random and patch-bay amounts. Small selectors stay as they are. In SEQ, `OCT-` + `KNOB 4` is nudge. |
| `PLAY` | Start or stop. Hold 0.7 s to stop and go back to step 1. |

## Playing

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| `STEP KEY` tap | The 16 white keys are the steps in SEQ, SYNTH, patch bay and kick pages. Each tap cycles OFF, TRIG, ACCENT. |
| `STEP KEY` hold + `KNOB` | Edit that step. In SYNTH pages it sets a parameter lock for the step (the value turns yellow, LOCK). On kick pages it edits the kick sequence. |
| `STEP KEY` hold + `REC` | Clear the step (SEQ or kick sequence). A kick step also goes back to pitch 0 and volume 100%. |
| MIDI note in | By USB, or by Bluetooth with the BLUETOOTH page on. A note plays the synth; note 36 triggers the kick as it is, without step pitch or volume. |

## Sequencer

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| `SEQ` page 1 | K1 pitch, K2 velocity, K3 gate (1-32 thirty-seconds of a step, or FULL), K4 ratchet (X1-X4). |
| `SEQ` page 2 | K1 condition (1:1, 1:2, 2:2, 1:3 ... 4:4, RND), K2 trig, K3 pattern length (1-16), K4 swing. |
| `SEQ` page 3 TEMPO | K1 BPM ±1 (it accelerates), K2 BPM ±10. The BPM is shown large. |
| `OCT-` hold + `KNOB 4` | NUDGE: shifts the whole sequence one step per click, wrapping inside the pattern length. Trig, pitch, velocity, condition, gate, ratchet and locks travel with their steps; the kick sequence stays. The counter shows the shift since you pressed `OCT-`. |

## Kick

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| `ARP` page 1 KICK SEQUENCE 1 | A sequence of its own, up to 16 steps, apart from the synth pattern. K1 trig (off, trig, accent), K2 pitch (-24 to +24 semitones), K3 volume (0-100%), K4 length (1-16). Step bars show the volume, accents in yellow. |
| `ARP` page 2 KICK SEQUENCE 2 | K1 condition, K2 ratchet (X1-X4), K4 nudge (shifts the kick sequence, wraps inside its length). |
| `ARP` pages 3-5 | 3: pitch and decay. 4: tone (click, drive, level, wave). 5: FM and noise. 9 presets. |
| Which step | The one you hold with a white key or, if none, the one `ALGORITHM` points to. |

## Sound

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| `EDIT` (SYNTH), 6 pages | Two oscillators with 16 types each (TRI SAW SQR PLS SIN PWM SSAW FM ORGAN VOWEL BRASS FOLD DIGI METAL NOISE BELL); filter and noise (LP4, LP2, BP, HP); envelopes; mixer (levels, sync, volume); modulation (envelope to VCO 1, VCO 2 and filter, accent); output (attack, drive, glide, tune). 16 presets. |
| Switches (SYNC, SLOPE ...) | A switch changes at the first click: right is ON, left is OFF. |
| Choices (waves, filter type, delay time, slices) | Each click moves to the next choice. |
| `LFO` (patch bay) | Eight virtual cables: source to target, depth -100 to +100, connected or not. Step keys work here too. 7 presets. |

## Effects

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| `FX`, 7 pages | COMPRESSOR, DELAY (ping-pong), REVERB (with width), STUTTER, DJ FILTER (high-pass with resonance, 12 or 24 dB), LIMITER and FX MIXER. 10 presets. |
| Order | Compressor, stutter, delay, reverb, DJ filter, limiter. |
| `KEY 1`-`KEY 6` | White keys 1-6 on the effect pages: compressor, delay, reverb, stutter, DJ filter, limiter on or off. |
| FX MIXER | K1 delay mix, K2 reverb mix, K3 DJ filter cutoff, K4 stutter slice (1/8 to 1/128). White keys: 1 delay, 2 reverb, 3 DJ filter, 4 stutter, 5 compressor, 6 limiter; white key 16 turns off delay, reverb, DJ filter and stutter together. |
| Stutter key | Tap to latch it. Hold 350 ms or more to stutter only while it is pressed. |

## Random

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| REC | Opens the RANDOM page of the page you are on (23 pages); press again to go back. Each knob has an ON/OFF switch and an amount (0-100%). |
| What a page says | IN or NOT IN OCT+ + HOME / REC: whether the general random draws this page. Under each bar, what that knob draws: a range (0-50), ANY for a choice, AUTO for a draw of its own. |
| RANDOM keys | White keys 1-4 switch a knob on or off, 5 all of them, 6 randomize this page, 7 the general random. |
| OCT+ + HOME | The general random: steps, synth sound and kick in one go. Cables, FX, scale and BPM stay. The resonance stays at 50 or under and the kick's level at 100 or under. |
| OCT+ + REC tap | The same general random with a wider draw, when you let go: the resonance up to 85, the kick louder. |
| OCT- + REC | Randomize only this page: whatever is switched on there (patch bay, FX and scale included). |
| OCT- + HOME | A melody: a voice that holds a pitch (no noise, sync or cables), one of six scales with the quantizer on (the root stays), a phrase built from a four-step figure and a kick on every beat. The kick's sound, the FX and the tempo stay. |
| OCT+ + REC hold 0.7 s | INIT: resets only the current pattern (HOLD TO INIT appears at 0.25 s). Tempo, other patterns, song and presets stay. |

## Patterns & song

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| GLO page 1 | Pattern grid: black keys are the banks A-K (11), white keys the patterns 1-16 of a bank (176 in all). While playing, the chosen pattern starts at the next bar. Actions: PLAY, COPY, PASTE, CLEAR, RAND. |
| GLO page 2 SONG | 32 slots. |
| GLO pages 3-4 PATTERN SOUND | Every pattern keeps its own sound (synth, kick, FX, patch bay, scale) and loads it when it plays. Page 3 shows the presets it came from (EDITED if changed). Page 4: scale preset, TO BANK and TO ALL (copy this sound to the bank or to all 176 patterns), RESET. |

## Scale

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| SCL, 2 pages | Root, scale (16: chromatic, major, minor, dorian, phrygian, lydian, mixolydian, locrian, harmonic and melodic minor, pentatonic major and minor, blues, whole tone, diminished, hirajoshi), octave, quantize on/off; transpose, fine tune, humanize, spread. Any key sets the root. Everything but the kick is quantized. 13 presets. |

## Saving & loading

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| SAVE page 1 PROJECTS | 4 slots (white keys 1-4). Pages 2-6: presets of SYNTH, KICK, FX, PATCH and SCALE (32 user slots each). |
| SAVE knobs | K1 item, K2 action (LOAD, SAVE, RENAME, CLEAR), K3 name cursor, K4 turn right to run it, or change the letter. |
| A project | Keeps the whole sound and banks A-C (48 patterns). Loading it replaces them. |
| Projects and backup | On the screen mirror page (Chrome or Edge, USB): Read the FM-1, then save a project to a file or replace one with a file, or save and load everything (sound, patterns, presets, projects). Stop the sequencer first; the FM-1 asks SEL to confirm, HOME to cancel. |
| Autosave | There is no save key. 1.5 s after your last change, with the sequencer stopped and nothing sounding; only what changed. While playing nothing is written (a flash write silences the audio for a moment). The SAVE action of a project or preset writes at once. |

## Sync, outputs & version

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| GLO page 6 MIDI | K1 MIDI SYNC, right ON, left OFF. On: the MIDI clock from USB, the MIDI jack or Bluetooth plays the steps, and its start, continue and stop run the sequencer; the header shows EXT and the clock's tempo. A step sounds 6.5 ms after its clock, always the same. No clock for half a second: the FM-1's own tempo. FM-FAM sends no clock. |
| GLO page 7 AUDIO | K1 USB audio ON/OFF (off: the computer records silence). K2 speaker + jack ON/OFF: one output, a plug switches the speaker off. K3 USB level: MASTER's, or FULL. Kept when you switch off: no sound? look here. |
| GLO page 8 VERSION | The firmware's version (0.2.5) and the audio's CPU load. |

## Bluetooth (experimental)

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| GLO page 5, or SELECT | The BLUETOOTH page. K1 ON/OFF (right is ON). K2, ALGORITHM or PRESETS scroll the devices (KEPT are remembered, NEW were found). K3 picks the action: SCAN, CONNECT, HANG UP, FORGET, OUTPUT, QUALITY, BT MIDI. K4 turned right runs it. |
| Pair headphones | Put them in pairing mode, K1 ON, SCAN, pick one with K2, CONNECT. Next time, CONNECT on the KEPT device (four are remembered). |
| OUTPUT | While linked, the sound goes to Bluetooth and the cable output is muted. OUTPUT switches it back. |
| QUALITY | HIGH, MID or LOW with each turn of K4. A read-out shows slots, bitpool and drops. |
| BT MIDI | Switches the Bluetooth MIDI announcement on or off (on by default). Notes sound as they do by USB. |
| Limits | The sound arrives 150-300 ms late: for listening, not for playing live. A device that accepts only one-slot packets sounds worse. After changing firmware, forget the FM-1 on the phone or speaker and pair again. Bluetooth is off at power-on unless you left it on. |

## Good to know

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| Random character | Each draw picks one: mostly low sounds, 15% of the time bright hi-hat sounds. Pitched steps use a minor pentatonic. |
| Screen mirror | Mirrors the screen and panel over USB (Chrome or Edge). The same page saves projects and backups, and installs versions. |
| Leave FM-FAM | Install any other firmware package with The Firmware Hub or M-UPGRADE. The package identity is FM-1_111. |

Community-made for the M-VAVE FM-1; not affiliated with or endorsed by M-VAVE. Everything on the screen is in English. FM-FAM is open source under the MPL-2.0: github.com/davidamomusic/fm-fam. All product names belong to their owners.